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Insider Threat Detection Model using Anomaly-Based Isolation Forest Algorithm

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ABSTRACT Insider attacks may inflict far greater damage to an organization than outsider threats since insiders are authorized users who are acquainted with the business's system, making detection harder. Many techniques to detecting insider threats have been developed, but they are neither flexible nor resilient owing to different obstacles (e.g., lack of real-world dataset and highly skewed class distribution of the available dataset), making insider threat detection an understudied research field. Previous techniques attempted to solve the dataset's imbalance issue by increasing or lowering the observations of the dataset's classes, however this might lead to underfitting and overfitting problems. We present an insider threat detection model that addresses the class imbalance problem at the algorithm level using anomaly-based techniques, as an enhancement over previous approaches. To limit the effect of skewed class distribution on insider threat detection, the Isolation Forest (IF) technique is used. The model is verified using the benchmarked CERT's insider threat dataset, which is significantly unbalanced, with a small number of malicious cases vs a large number of non-malicious instances. Several contamination ratios of IF's parameters are used to verify the model's performance throughout a range of anomaly scores. The experimental findings reveal that the suggested model handles the dataset class imbalance problem with an accuracy score of 98%. The findings are compared to the baseline technique to demonstrate how the proposed model enhances detection performance and addresses the problem of data imbalance. The findings indicate the usefulness of the suggested approach for identifying insider threats when compared to previous studies.

INDEX TERMS Anomaly detection, Dataset Imbalance Issue, Insider threat detection, Isolation Forest, Machine learning.

I. INTRODUCTION

Insider threats represent 25% of all attacked organizations and their frequency is growing [1]. The term “insider threat” refers to threats posed by an insider, either intentionally or unintentionally, who has access to the organization's network system and data to jeopardize the confidentiality, integrity, availability, or physical well-being of the organization's information, information systems, or workforce [1]. Insider attacks are considered the most difficult type of attacks to detect and the costliest. According to cybersecurity reports, insider threats affect 42% of U.S. federal agencies and 53% of organizations annually [2], [3].

Existing systems for identifying insider threats function well; but, typical Machine Learning (ML) algorithms do not

utilize all of the data from user behavior. This is due to the data's complexity, high dimensionality, sparsity, and heterogeneity. In ML algorithms, it is usually assumed that data are balanced. However, such unbalanced data results in high accuracy in recognizing the majority class and poor accuracy in detecting the minority class. In the event of an insider threat, this conclusion is unsuitable since the minority class is crucial in detection [4], [5].

Researchers have proposed many ways, the most prominent and general of which is to rebalance the class distribution via data augmentation methodologies [6]. The simplest way is to oversample the minority class by randomly repeating its instances. However, this strategy yields no new information. In order to address this problem, the Synthetic Minority

Oversampling Technique (SMOTE) approach [7] generates synthetic samples based on samples from the minority class using a linear process. Unfortunately, these techniques only consider local information, resulting in unitary synthetic samples and overfitting.

As a result, in order to address the issue of dataset class imbalance in insider threat detection, this article proposes an insider threat detection model based on a machine learning anomaly detection approach. So, rather than addressing the imbalanced classes of the dataset through duplication (oversampling) and removal (under-sampling) of dataset samples, as several related works have done, our model addresses the imbalance issue at the algorithm level rather than the data level. To that purpose, the following contributions are hoped for:

- It focuses on identifying insider data leakage attacks carried out by a malevolent insider who then departs a company.
- It employs well-known anomaly-based IF machine learning algorithm to solve the CERT dataset's excessively skewed class distribution using outlier identification mechanisms known as One-class classification algorithms
- It evaluates the minority class of the dataset using different contamination ratios to simulate various outliers/anomalies for the aim of developing a generalized insider threat detection system that can be applied on other anomaly detection applications.
- It employs two approaches to demonstrate the enhanced accuracy of the proposed model: the baseline, in which the supervised machine learning algorithms are executed on the unbalanced dataset, and the anomaly detection techniques.

The rest of the paper is organized as follows. In Section 2, we discussed the related works by presenting a high-level summary of the most important aspects (Methodology, Dataset, Results, etc.) of the related work. Section 3 explains the approach and core components of the proposed model. The experimental settings and results for assessing the performance of the proposed anomaly-based IF machine learning model. It also discusses and compares the results of the proposed method with the baseline and related work. Finally, Section 5 concludes the work and proposes future research options.

II. RELATED WORK

Government agencies and cybersecurity firms are facing difficulties in detecting insiders. Even for the research community, detecting insider threats is a difficult problem. Common guidelines have been issued by The Computer Emergency Response Team (CERT) Insider Threat Center and the U.S. National Insider Threat Task Force to protect organizations from insider threat [1], [8]. A number of 20 practices are described in [1] to prevent and detect insider

threats from occurring in organizations as well as examples of organizations that failed to implement these practices.

Detection approaches of insider threat have been overviewed from different angles in recent surveys [9], [10], [11] and [12]. Malware and advanced persistent threat issues related to insider threat are discussed in [9]. The authors in [10] present a structural taxonomy and novel categorization of studies on insider threats. The study in [11] demonstrates the characterization and comparison of several empirical strategies for detecting insider threat with regard to ten major variables (e.g., datasets, feature domains, classification algorithms, simulated scenarios, performance and accuracy metrics, and so on). The survey in [12] examined insider threat detection methodologies from several perspectives, including detection techniques, machine learning methods, datasets, analyzed behaviors, and performance indicators. There was also a categorization of current insider threat kinds, access strategies, levels, reasons, insider reporting, security resources, and so on. Also mentioned were the issues associated with insider threat, as well as some solutions for overcoming them.

According to the authors of [10], there is a small body of work in the insider threat literature where deep learning methods have been used to balance the unbalanced class samples. Oversampling is employed in [13] to balance the class imbalance, which is subsequently followed by an ensemble of one-class classifiers. The authors advocated oversampling of false-positive instances using a Local Outlier Factor (LOF) to produce synthetic data points. The notion of insider threat has been explained in [14] and challenges of classification resulting from the lack of real-world data has also been discussed. The study simply offers a conceptual framework, and there are no specifics provided about the technical aspect for properly managing imbalanced data.

Three separate binary classification problems were taken into consideration in the scenario-based classification in the study [15]. A time series categorization of user activity was offered as a strategy for detecting insider threats. It employed a cost-sensitive data adjustment strategy to categorize the unbalanced real-world insider threat data, which only comprised a small number of malicious cases, by randomly under-sampling the non-malicious class instances. It compared the performance of a two-layered deep autoencoder neural network classifier to that of random forest and multilayer perceptron, two other popular classifiers. Both deep autoencoder and random forest classifiers categorized the data-adjusted time-series feature set with good accuracy, recall, and f-score. The multilayer perceptron has a poorer accuracy and f-score while having a greater recall than the other two classifiers.

Although insider threat analysis has been researched for a long time, owing to a lack of real-world data, it has been difficult to adequately study appropriate remedies. One of the most serious issues confronting the insider threat research community is a scarcity of real-world datasets. It is garnering

greater attention these days due to an increase in assault frequency and the introduction of promising data analysis techniques. The CERT benchmarked insider threat dataset [16] is significantly lopsided, with a minority class having a small number of positive (malicious) incidents compared to a majority class having a huge number of negative cases (non-malicious). As a consequence, building a machine learning model on a massively skewed dataset presents several obstacles (e.g., the skewness of the results toward the majority class). Except for a few papers that provided some insight into the topic, this issue in the context of insider threat detection has received little attention in the literature.

A deep adversarial insider threat detection (DAITD) system was proposed in [17] by using Generative Adversarial Networks (GAN) to approximate the real anomalous behavior distribution. The LSTM-Autoencoder, which was employed as a user behavior encoder, was used to construct the anomalous user behavior representations from the anomalous behavior data (minority class). The GAN's generator was then used to represent the actual anomalous behavior distribution, while the GAN's discriminator was utilized to assess whether or not the synthetic sample from the generator was genuine. The authors generated the data utilizing a GAN with fully connected layers. The performance of the data-adjusted samples is evaluated using tree-based classifier models, which are also susceptible to overfitting but may offer very high-

performance ratings. The performance looks to be extraordinary when compared to all other models, yet the technique is significantly reliant on assumptions. The two challenging characteristics of GAN, vanilla convergence and mode collapse, are not covered in detail. The evaluation of generated data from the tests, both qualitatively and quantitatively, is very ambiguous. Furthermore, it is questionable if the data produced by generative models is robust.

In general, the strategies that are now being deployed make use of techniques that include machine learning in order to detect insider threats. When the dataset that is being utilized is highly skewed, machine learning algorithms may be an effective method for identifying insider threats; nevertheless, these algorithms have the potential to be biased and provide less accurate results. The problem of an imbalanced insider threat dataset has not been addressed in the literature in a way that is either obvious or adequate. The existing work has a limitation in that it addresses the imbalanced dataset by either oversampling (duplicating the minority class instances) or under sampling (removing the majority class instances), both of which are impractical because they manipulate the raw dataset and produce no real-world dataset representation. This is a limitation that the existing work has. A brief overview of the most relevant works is shown in Table1.

TABLE 1:
A SUMMARY OF THE MOST RELEVANT WORK.

Ref	Year	Method	Pros	Cons
[18]	2017	Deep Belief Network	Appropriate for applications requiring simple processing of features across multiple domains	Produces a high percentage of false positives and has trouble dealing with temporal data.
[19]	2018	Convolutional Neural Network	When paired with LSTM, enables the processing of feature matrices to take place at a faster rate.	Without LSTM, it is unable to deal with temporal information.
[20]	2018	Autoencoder	Organization-level solutions that operate autonomously and generate fewer false positives.	Irresponsible with time-based information.
[21]	2021	Logistic regression (LR), Decision tree (DT), Random forest (RF), Naïve Bayes (NB), k-nearest neighbors (k-NN) and Kernel support vector machines (KSVM).	There are three techniques that are utilized here: label encoding, one-hot encoding, and SMOTE. Addresses any potential problems with prejudice and class inequality.	As opposed to using a real dataset, the model is tested on a simulated one.

To address the issue of unbalanced categorization, this study proposes a machine learning-based anomaly detection solution. This method of anomaly identification based on

machine learning is essential for rectifying the issue of uneven datasets. This is made feasible by using the IF machine learning algorithm, which is based on the detection of

anomalies. This method can determine if a potential new instance is benign or (anomalous/malicious). Anomaly detection has been proven to be particularly useful when the minority class is extremely disorganized, consisting largely of noisy instances or small disjoints. Extensive testing has shown this to be the case [22]. Our experimental results show that the proposed model enhances insider threat detection performance and provides a better answer to the imbalance problem of insider threat detection. The proposed model is compared with the baseline and other models used to detect insider threats in the IV. EXPERIMENTAL EVALUATION section.

III. METHODOLOGY

The goal of this study is to evaluate the performance of a machine learning-based anomaly detection model in detecting

insider threats throughout an organization's network. To that purpose, we apply the processes shown in Fig. 1. This section illustrates the approach and key components of the proposed model. The dataset and preparation techniques are shown in Section A. Section B discusses how to address the dataset imbalance issue. To assess the model's performance, we first use supervised machine learning approaches as a baseline (XGBoost, Random Forest, Decision Tree, and K-Nearest Neighbor). The anomaly-based IF machine learning method is then implemented, and the results are compared to the baseline as well as previous work for performance evaluation. An overview of the proposed model is presented in Fig. 1.

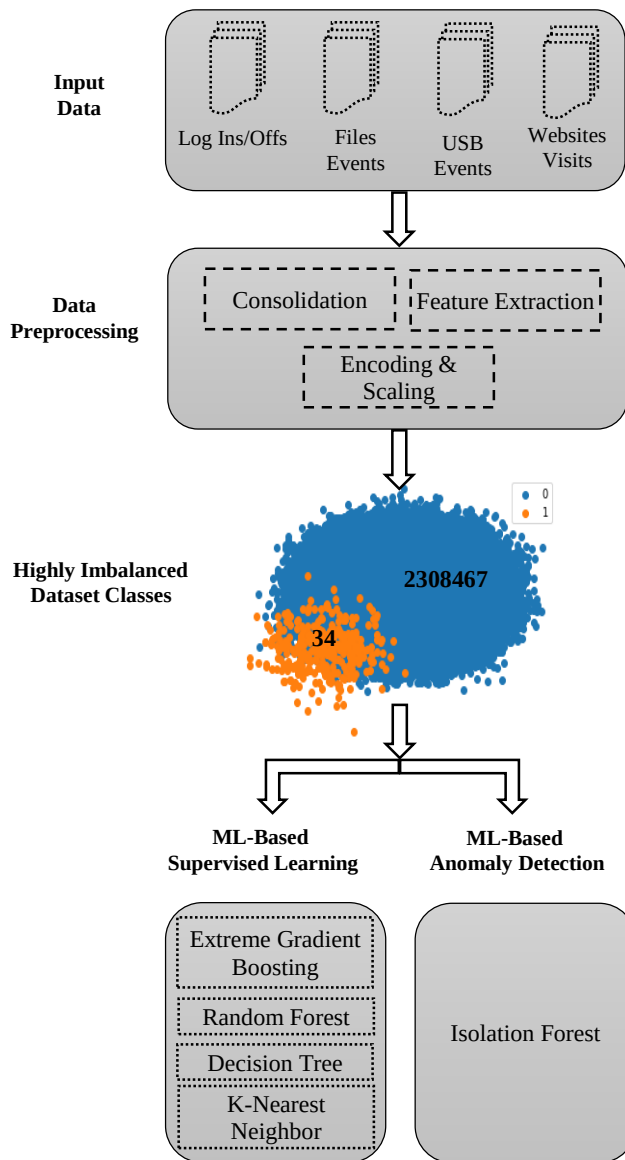


FIGURE 1. An overview of the anomaly detection approach based on machine learning.

The proposed insider threat detection model using supervised learning algorithms as a baseline and the Isolation Forest algorithm for anomaly-based detection involves several steps, including data preparation, feature engineering, model training, and evaluation. Below is a high-level mathematical model:

1. Problem Definition:

- Let D be the CERT dataset containing information about user activities and incidents.
- Let X represent the feature matrix with dimensions (m, n) , where m is the number of samples, and n is the number of features.
- Let Y represent the target variable indicating insider threat labels (0 for non-threat and 1 for threat).

2. Data Preprocessing:

- Perform data preprocessing to prepare the dataset for modeling:
 - Handle missing data.
 - Encode categorical variables into numerical representations.
 - Normalize or standardize features.
- Split the dataset into training (D_{train}) and testing (D_{test}) sets for model evaluation.

3. Supervised Learning (Baseline):

- Choose a supervised learning algorithm, such as logistic regression, decision trees, or random forests, for the baseline model.
- Train the model: $M_{\text{baseline}} = \text{Train}(X_{\text{train}}, Y_{\text{train}})$
- Evaluate the model's performance on the test set: $P_{\text{baseline}} = \text{Evaluate}(M_{\text{baseline}}, X_{\text{test}}, Y_{\text{test}})$

4. Isolation Forest (Anomaly-Based Detection):

- Implement the Isolation Forest algorithm for anomaly-based detection:
 - Let $IF(X, \psi)$ represent the Isolation Forest model, where X is the feature matrix, and ψ are hyper-parameters.
 - Train the model: $IF(X_{\text{train}}, \psi) \rightarrow IF_{\text{anomaly}}$
 - Predict anomalies on the test set: $S_{\text{anomaly}} = \text{Predict}(IF_{\text{anomaly}}, X_{\text{test}})$

5. Model Evaluation:

Evaluate both the baseline supervised learning model and the Isolation Forest model using appropriate metrics Accuracy, Detection Rate and F-score.

- For the supervised model: $P_{\text{baseline}} = \text{Evaluate}(M_{\text{baseline}}, X_{\text{test}}, Y_{\text{test}})$
- For the Isolation Forest: $P_{\text{anomaly}} = \text{Evaluate}(S_{\text{anomaly}}, Y_{\text{test}})$

6. Model Comparison:

- Compare the performance of the supervised learning baseline model (P_{baseline}) with the anomaly-based Isolation Forest model (P_{anomaly}).

- Select the model that provides better results in terms of the Accuracy, Detection Rate and F-score evaluation metrics.

The aforementioned mathematical model of the proposed work provides a structured approach to building an insider threat detection system using the Isolation Forest algorithm. The details of the model and the actual implementation using python programming language and appropriate libraries and tools for machine learning and data analysis are presented in the following sections.

A. Dataset and Pre-processing

Dataset preparation and preprocessing are crucial steps for cybersecurity solutions, as they enable an efficient implementation of machine learning techniques as well as supporting security analysts for taking the right decisions. Finding an actual dataset to validate the suggested insider threat detection systems is one of the most difficult tasks for the cybersecurity research community. The privacy problem is the primary reason for the absence of a real dataset in insider threat detection studies. The publicly available and widely applied synthetic dataset for validating an insider threat detection system is the CERT's dataset [16] provided by Carnegie Mellon University. It has been used widely in several studies, such as [23], [24], [25], [26] and [27]. The description of how the dataset was created is presented in [28].

In this study, we utilized the CERT r4.2 datasets which is the most widely employed dataset in the literature. Thousand employees are simulated in the dataset, 7% of them are malicious insiders. This allows us greater flexibility when conducting experiments for anomaly detection and helps us learn how well the model performed. The dataset provides log data for 1000 users' activities spanning for 500 days and containing logons, HTTP, email, file, and removable devices activity logs. An overview of the dataset files is shown in Table 2.

TABLE 2

AN OVERVIEW OF THE UTILIZED INSIDER THREAT DATASET FILES.

File	Description
Logon	The two forms of activities that are noted in logon.csv file are logons and logoffs. It contains information on computer use, such as session length, frequency of sessions during and after working hours, and other parameters. This data sheds light on the access patterns of the user.
File	Logs of users' files operations is contained in the file.csv file. The file access pattern of the user is examined using this information.
Device	The actions of this file, including device connection and disconnection, serve to show the use of removable drives on PCs. The user's daily access information to external devices is included in the Device.csv log file. It shows

File	Description
	when a user accesses any device, including during and after working hours.
Email	Logs of email activities of users are included in the email log file. Whether an internal or external sender or recipient of emails was involved is also mentioned.
HTTP	The user's browsing history is detailed in the http.csv file. All HTTP records contain the following information: user, PC, URL, and web page content. The HTTP log file stores all URLs to websites that users have visited while connected to the organization network.

Several data preprocessing steps are conducted on the raw dataset. Then, the matrix of features representing the attack scenario is fed into the anomaly detection algorithms. Firstly, we combine the files of insiders' activity logs that involve particular features of the insider threat scenario of interest. Secondly, we apply dataset refining, feature extraction, encoding, and scaling. Finally, the matrix of features that represent normal and malicious instances of insiders is generated to train and validate the applied machine learning algorithms.

The logs of insiders' activities, which were originally involved in several files are consolidated into one homogeneous file. Some data fields may have contaminants while gathering user activity logs from various sensors placed throughout an organization (e.g., missing data). This is typical as the dataset is gathered by a number of sensors dispersed around the company, some of which may not always function as intended. The combined files are verified to make sure there are no null values because the categorization process places a high priority on data quality.

The attributes that indicate a scenario involving an insider data leakage attack are extracted once the dataset files have been integrated and refined. Because the inclusion of noisy data may impair the performance of machine learning algorithms, it is impractical to include all of the data in the dataset. The selection of the most pertinent features affects the performance of the machine learning algorithms. So, the relevant features, Vectors (logon, logoff, device, http), Timestamps, User IDs, Events (e.g., login/ logoff, USB connect/disconnect, http visits, etc.) and Targets (normal or malicious).

The observations to detect an insider threat are selected to reflect the threat scenario. The threat scenario studied in this paper is that "a user who does not usually uses a USB device begins to use it more after working hours and leaves the organization thereafter". Thus, some key features that reflect the attack scenario include (# of log in/off outside of working hours, # of USB devices usages, and # of WikiLeaks website visits). Numerical vectors, also known as data instances that summarize user actions, are produced through feature

extraction on the aggregated data. Each vector is made up of user input, the majority of which is category data stored in numerical representation for the purpose of giving machine learning algorithms context.

B. ADDRESSING DATASET IMBALANCE ISSUE

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1) SUPERVISED LEARNING

Supervised machine learning approaches such as Decision Tree, XGBoost, Random Forest, and K-Nearest Neighbor

generate data categorization models from labeled training data. This section describes the supervised machine learning approaches that we employed in our approach. The Decision Tree (DT) algorithm is one of the most widely used categorization methods [29]. A decision tree is analyzed using the divide and conquer method, with decision nodes establishing tests on one characteristic and leaf nodes reflecting class values. A set of rules governs every route from a leaf node to the root node. The single most essential performance component is classification error, defined as the fraction of misclassified cases [30]. If there are multiple classes in the training set, there are often too many levels and branches in a decision tree. As a result, as the number of classes grows, the classification's accuracy declines. The Decision Tree algorithm in insider threat detection works as follows:

$$F(x) = \text{DecisionTree}(x) \quad (1)$$

- $F(x)$: The final prediction for the sample x (0 for non-threat and 1 for threat).
- $\text{DecisionTree}(x)$: A function that represents the decision tree's prediction for the sample x .

In insider threat detection using a Decision Tree algorithm, the final prediction $F(x)$ for a given sample x is determined directly by the decision tree. The decision tree processes the features of the sample and follows its branches and leaves to make a prediction, classifying the sample as a threat or non-threat based on the learned rules and splits within the tree.

A popular boosting model that delivers great job for resolving numerous classification issues is the Extreme Gradient Boosting (XGB) method [31]. In addition to having a low computing cost with automated learning in the splitting process, it offers a novel tree searching technique for avoiding an overfitting problem, particularly with highly sparse data. In the XGB technique, decision trees are employed to generate an ensemble of classifiers [32]. It is assumed that a gradient boosting strategy will be utilized, with the optimization of a differentiable loss function resulting in the combination of basic decision trees into a robust ensemble. Additionally, in boosting, each decision tree output is combined linearly to provide probabilities that are used to forecast class labels using the logistic transformation. It offers advantages over conventional gradient boosting techniques to enable a tree boosting system that is very scalable. Numerous implementations of the XGB algorithm have been effective in data mining tasks [33] and [34]. We employ XGB in this investigation to evaluate its ability to recognize insider threats due to its widespread use. The foundation of the XGB algorithm is further described in [31]. The following is an equation for applying the Extreme Gradient Boosting (XGBoost) algorithm in insider threat detection:

$$F(x) = \sum_{t=1}^T f_t(x) \quad (2)$$

- $F(x)$: The final prediction for the sample x (0 for non-threat and 1 for threat).
- T : The total number of decision trees in the XGBoost ensemble.
- $f_t(x)$: The prediction of the t -th decision tree for the sample x .

In insider threat detection using XGBoost, multiple decision trees are trained and combined to make a final prediction. Each decision tree $f_t(x)$ provides its prediction for the given sample x , and the final prediction $F(x)$ is the sum of these individual tree predictions. The ensemble output is used to classify whether the sample represents a threat or not.

The Random Forest is a machine learning technique that is used to improve the accuracy of numerous categorization tasks (RF). It is made up of several decision trees with a minimal number of nodes and a variety of attributes for dividing each node. The most suitable node for splitting is chosen at random when building each unique tree. The creation of forests that may be retained for later use is one of the numerous benefits of the RF algorithm [35]. Additionally, RF resolves the overfitting problem [36], and automatically generates the variable importance [37]. The same procedure is used to construct a large number of decision trees. The properties and design of the decision tree nodes are decided based on the subclass of instances used to produce each decision tree and the subclass of attributes used to generate each decision node. These qualities have been formally proved to avoid over-learning and to improve the robustness of RF predictions [38]. However, the RF algorithm might suffer from a decrease in classification tasks' accuracy because it creates a number of noisy trees [39]. The following is a simple equation for applying the Random Forest algorithm in insider threat detection:

$$F(x) = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (3)$$

- $F(x)$: The final prediction for the sample x (0 for non-threat and 1 for threat).
- T : The total number of decision trees in the Random Forest ensemble.
- $f_t(x)$: The prediction of the t -th decision tree for the sample x .

In insider threat detection using Random Forest, multiple decision trees are trained and combined to make a final prediction. Each decision tree $f_t(x)$ provides its prediction for the given sample x , and the final prediction $F(x)$ is the average (or majority vote) of these individual tree predictions. The ensemble output is used to classify whether the sample represents an insider threat or not.

A non-parametric machine learning approach called K-Nearest Neighbor (KNN) [40] examines how similar unlabeled instances are to all the data training set. The training data instances are vectors in a multidimensional feature space, and each instance is assigned a class name. An infinite vector is classed by giving the frequent label to the closest k training examples, where k is a user-defined constant number. The

KNN employs the widely used distance metric known as Euclidean distance. The equation for applying the K-Nearest Neighbors (K-NN) algorithm in insider threat detection is as follows:

$$F(x) = \text{MajorityVote}(k, \text{Neighbors}(x)) \quad (4)$$

- $F(x)$: The final prediction for the sample x (0 for non-threat and 1 for threat).
- k : The number of nearest neighbors to consider.
- $\text{Neighbors}(x)$: The set of k nearest neighbors to the sample x .
- $\text{MajorityVote}(k, \text{Neighbors}(x))$: A function that returns the majority class label among the k nearest neighbors.

However, the long training time, high computing cost and the possible bias detection results on highly imbalanced datasets are some of the limitations of employing the supervised machine learning algorithms [41]. Thus, we apply the anomaly detection to address the highly imbalanced insider threat dataset while detecting data leakage attack incidents for the aim of achieving more reliable insider threat detection results.

2) ANOMALY DETECTION

Anomalous data are unusual in contrast to normal data, and they do not fit neatly into the dataset's distribution. Anomaly detection is the technique of spotting outliers in data. Anomaly detection strategies are supplied by the field of machine learning. It refers to unsupervised machine learning algorithms that employ normal examples to identify novel situations as normal or abnormal. Such strategies may be applied for binary classification when the distribution of dataset classes is highly skewed. They are trained on the dataset majority class's input instances and then tested on a test dataset. The malicious observations dominate the normal observations in the CERT insider threat dataset utilized in this research. As a consequence of the occurrence of such extreme class distribution, the minority class can be counted as outliers, leading the classification results to skew.

Rather than traditional supervised learning algorithms, the IF [42] constructs decision trees to detect anomalies. The decision tree partitions are produced by randomly selecting a feature from the training dataset, followed by a random split value between the feature's minimum and maximum value. The IF's core tenet is that outliers are abnormal instances that deviate from the pattern of the majority of dataset instances. These outliers will be located nearer to the tree's root due to the random partitioning, resulting in trees with shorter paths. An instance is an outlier if it is located along a tree's shorter branch. So, using the following equation [43], an anomaly/outlier score is calculated:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (5)$$

where $h(x)$ is the instance's x 's path length. The $c(n)$ is the typical length of a binary tree search that is unsuccessful. The parameter n determines how many external nodes there are, and $E(h(x))$ is the average value of $h(x)$ over a group of trees in an isolation forest [42]. Each instance is given a score; as a result, if $S(x, n)$ is close to 1, it denotes an anomaly or outlier. A score of less than 0.5 implies a normal instance in all other cases. The equation for applying the Isolation Forest algorithm in anomaly-based insider threat detection is as follows:

$$S(x) = \frac{c(n) \cdot h(x)}{T} \quad (6)$$

- $S(x)$: The anomaly score for the sample x .
- $c(n)$: A normalization factor based on the number of samples (n).
- $h(x)$: The path length of the sample x in the isolation tree.
- T : The total number of isolation trees in the Isolation Forest ensemble.

In anomaly-based insider threat detection using the Isolation Forest algorithm, each sample's anomaly score $S(x)$ is calculated based on the path length $h(x)$ of that sample in an individual isolation tree. The anomaly score is then normalized by a factor $c(n)$ and averaged across all the trees in the ensemble (T). Lower anomaly scores indicate potential insider threats, while higher scores suggest normal behavior. An appropriate threshold can be set to classify samples as threats or non-threats based on these anomaly scores.

According to Fei Tony Liu et al. [42], the majority of current model-based techniques for anomaly identification first create a profile of normal instances before classifying instances that deviate from the normal profile as anomalies.. The approach used by Isolation Forest, on the other hand, explicitly isolates anomalies rather than profiling normal points, by using its quantitative characteristics. They have demonstrated how to efficiently create a tree structure to isolate each instance. This is accomplished by randomly creating hyperplanes that divide instances from each other in multiple dimensions. The anomaly score of an instance is correlated with the number of hyperplanes required to separate out the instance.

Due to their proximity to the tree's root, outliers are easily detected by the IF. If the dataset is free of outliers and there are many irrelevant features in high dimensional problems, it performs well [42]. Because of this, the IF algorithm works well with our dataset's high imbalance. It can also be used with just regular data without suffering any noticeable performance losses [42], which is extremely helpful in insider threat detection.

C. Performance metrics

In this section, we provide a summary of the evaluation metrics Accuracy, Detection Rate/Recall, and F-score in the context of insider threat detection:

•**Accuracy:** Accuracy measures the proportion of correctly classified instances (both true positives and true negatives) among all instances in a dataset. In insider threat detection, accuracy reflects the overall correctness of the model's predictions [20]. Accuracy is a commonly used metric to assess the overall performance of insider threat detection models. It provides a general measure of how well the model is classifying both legitimate and malicious activities.

•**Detection Rate(DR):** DR, also known as Recall or True Positive Rate, measures the proportion of actual insider threats (true positives) correctly identified by a model. In insider threat detection, recall indicates how well the model detects malicious activities while minimizing false negatives [21]. Recall is a critical metric in insider threat detection as it quantifies the model's ability to identify true insider threats, which is a primary concern in security applications.

•**F-score:** The F-score, particularly the F1-score, is a harmonic mean of Precision and Recall. It provides a balanced measure of a model's performance, taking into account both false positives and false negatives. In insider threat detection, it gauges the trade-off between precision and recall. The F-score is valuable in insider threat detection when there is an imbalance between normal and threat instances. It helps assess a model's ability to correctly classify threats while maintaining a reasonable level of precision [44].

These evaluation metrics play a crucial role in assessing the effectiveness of insider threat detection models, helping security practitioners and researchers make informed decisions about the performance of their systems in real-world security scenarios.

IV. EXPERIMENTAL EVALUATION

This section presents the experimental settings and results for assessing the performance of the proposed anomaly-based IF machine learning model. First, the preprocessing steps (section A) are carried out using the benchmarked CERT r4.2 insider threat dataset [16]. Second, the supervised machine learning algorithms are applied as a baseline. Third, the proposed anomaly-based IF machine learning algorithm is applied to address class imbalance issue inherent in the CERT dataset. Finally, the analysis of experimental results as well as the comparison with the baseline and existing work are presented.

The implementation is done utilizing Python 3.10.4 programming language with Scikit-learn libraries [45]. A summary of the experimental tools is outlined in Table 3.

TABLE 3
THE UTILIZED TOOLS.

Tool	Specification
OS	Microsoft Windows 10
Processor	Intel(R) Core (TM) i5-4590S 64-bit processor, CPU 3.00GHz
RAM	8 GB
Python	3.10.4
Scikit-learn	1.1.0

The Google Colaboratory (known as Colab) is a cloud service based on Jupyter notebooks hosted on the Google Cloud platform. It provides several advantages by enabling code to solve different challenges, which triggers different organizations in incorporating Google Colab as an Information and Communications Technology in Data Science [46]. Thus, we employ Google Cloud in our experiments. We implement several preprocessing steps (consolidating, addressing missing data, refining, feature extraction and encoding) on the dataset as briefed in section A. Then, the data feature matrix is fed into machine learning models.

Beyond the accuracy, the most commonly used metrics (Recall and F-score) are applied to evaluate the effectiveness of the proposed model. The Recall known as Detection Rate or True Positive Rate measures the proportion of insider instances that are detected correctly. The F-score is also adopted for its expressive power and it is a popular metric for imbalanced classification [47]. This paper addresses the highly imbalanced class distribution of the CERT dataset for insider threat detection. The class distribution that confirms the extremely imbalance issue of the dataset is summarized in Fig. 2.

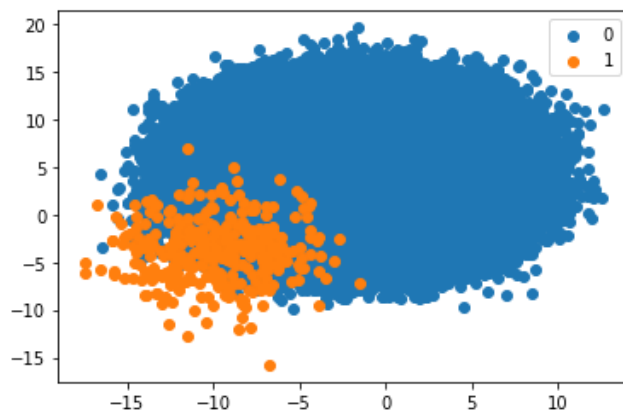


FIGURE 2. The class distribution of the CERT dataset.

The scatter plot shows the imbalanced insider threat classification problem. As it is shown in Fig. 2, the scatter plot is created as points colored by their class label. The majority class (blue) represents the normal events of insiders while the minority class (orange) represents the malicious events of

insiders. A summary of the dataset class distribution is shown in Table 4.

TABLE 4
THE CLASS DISTRIBUTION OF THE DATASET.

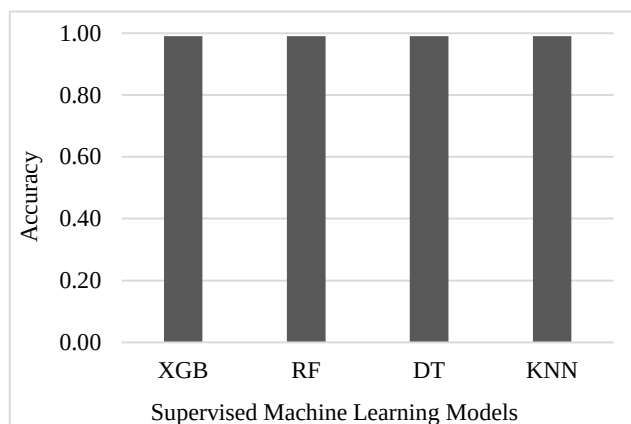
	Class 0	Class 1
Representative	Normal	Malicious
Number of events	2308467	346
Percentage	99.985%	0.015%

As it is shown in the Table 4, the dataset is extremely imbalanced, so our approach is to apply the supervised machine learning algorithms on such an imbalanced dataset distribution as a baseline.

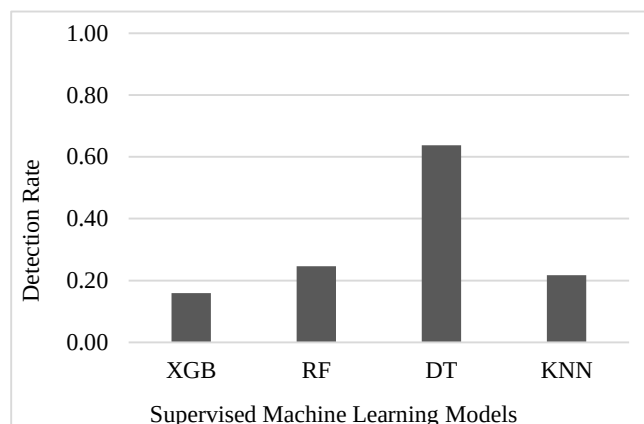
A. SUPERVISED MACHINE LEARNING (BASELINE)

The goal of this research is to create an anomaly-based machine learning system for insider threat detection to address

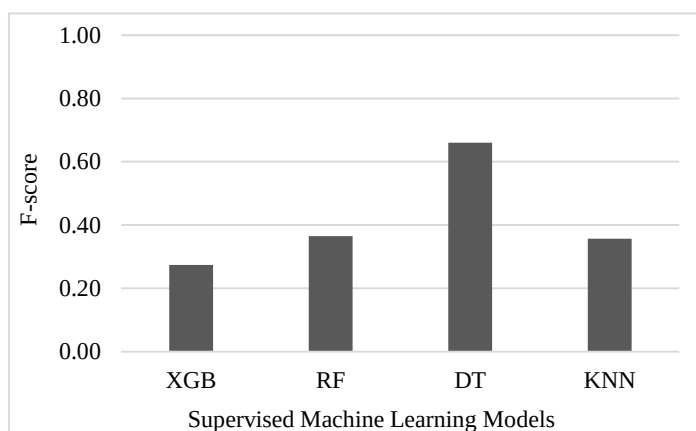
the highly imbalanced issue in the dataset. In this section, the supervised machine learning algorithms are applied to detect malicious events that might be carried out by an insider. The dataset is used to train and fit a mathematical or statistical machine learning model. The model learns the dataset's pattern for the aim of separating normal events from malicious ones on unseen data in the evaluation process. The dataset is split into 80% for training and 20% for testing. The dataset used to fit the model is referred to as training data which consists of 1846773 normal instances and 277 malicious instances. The trained model is then utilized to predict unseen data referred to as test data which consists of 461694 normal instances and 69 malicious instances. In this way, the built model is able to detect new observations without human intervention. An overview of the applied machine learning algorithms (XGB, RF, DT and KNN) is summarized above. The results of applying the supervised machine learning algorithms are presented in Fig. 3.



(a)



(b)



(c)

FIGURE 3. Subfigures (a), (b) and (c) are the accuracy, detection rate and F-score results of applying the supervised machine learning models (Baseline).

XGB, RF, DT, and KNN are the four machine learning algorithms used in the baseline experiments, and their performance is evaluated based on the accuracy, recall, and f-score criteria. Figure 3(a) demonstrates that the combined detection accuracy of all used algorithms is 99%. Results are unreliable/misleading with such a high accuracy score because of the significantly uneven issue of dataset classifications (see Fig. 2 and Table 4). The DT method achieves a 64% detection rate and an f-score of 66% with regards to additional assessment criteria (subfigures (b) and (c) of Fig. 3). To put it another way, with 20 FP and 25 FN, the DT algorithm is not sensitive enough to detect instances of insider activity. When compared to other algorithms in use, the DT method achieves the best F-score (66%), indicating that it is more effective at detecting malicious activity while also producing fewer false positives. The RF and KNN algorithms obtain nearly the same results with (Detection Rate of 25% and F-score of 37%) and (Detection Rate of 22% and F-score of 36%), respectively. While the RF algorithm obtains 7 FP instances, the KNN algorithm receives zero, making the KNN algorithm the clear winner. With a detection rate of 16% and an F-score of 27%, the XGB algorithm achieves the highest FN (58 instances) of all the algorithms tested. Table 4 summarizes the outcomes of using the four supervised machine learning methods and should help clear things up.

Analyzing and discussing the results of supervised machine learning algorithms (XGBoost, Random Forest, Decision Tree, and K-Nearest Neighbors) with the provided F1-score results (0.27, 0.37, 0.66, and 0.36, respectively) requires understanding the performance of each algorithm in the context of insider threat detection. The F1-score is a harmonic mean of precision and recall, providing a balanced measure of a model's accuracy. Higher F1-scores indicate better overall model performance. Decision Trees have demonstrated the highest F1-score of 0.66, which indicates a good balance between precision and recall. This result suggests that the Decision Tree model has successfully learned the patterns in the data and can effectively classify insider threats while minimizing false positives and false negatives. Decision Trees are known for their interpretability, making them valuable for understanding the features that contribute to threat detection.

Random Forests, an ensemble of Decision Trees, have an F1-score of 0.37. While Random Forests often improve upon the performance of individual Decision Trees by reducing overfitting, this result indicates that the model may still have room for improvement. Hyperparameter tuning and feature selection can be explored to enhance the Random Forest model's performance further.

K-Nearest Neighbors (KNN) achieved an F1-score of 0.36. KNN is a distance-based algorithm that can be sensitive to the

choice of hyper-parameters, such as the number of neighbors (k). The relatively lower F1-score suggests that KNN might not be the best-performing algorithm for this specific insider threat detection task.

XGBoost, an ensemble learning method, achieved the lowest F1-score of 0.27 among the evaluated models. While XGBoost is a powerful algorithm, this result indicates that it may not have been properly configured or that other algorithms were more suitable for the dataset and problem at hand. Hyper-parameter tuning and feature engineering could potentially improve XGBoost's performance.

In summary, The Decision Tree model performed the best, with a relatively high F1-score, indicating good precision and recall. Random Forest and K-Nearest Neighbors had F1-scores lower than the Decision Tree but might benefit from further optimization. XGBoost, despite its capabilities, had the lowest F1-score, suggesting that it may require adjustments to its hyper-parameters or feature preprocessing to achieve better results. A summary of obtained results by the four supervised machine learning models is showed in Table 5.

TABLE 5

THE RESULT DETAILS OF APPLYING THE SUPERVISED MACHINE LEARNING MODELS.

ML Algorithms	Accuracy	Detection Rate	F-score
XGB	0.99	0.16	0.27
RF	0.99	0.25	0.37
DT	0.99	0.64	0.66
KNN	0.99	0.22	0.36

B. ANOMALY DETECTION

The IF is a tree-based anomaly detection algorithm that detects anomalies based on the concept of isolation without using any distance or density measure [48]. It models normal data instances in such a way that instances that are both few in number and different in the feature space are isolated as anomalies. It is implemented on the dataset utilizing the scikit-learn library [45] by setting hyperparameters of the algorithm as; the “n_estimators” parameter for the number of trees to create, and the “contamination” parameter for defining the number of outliers in the dataset. The contamination parameter represents the percentage of positive instances to negative instances or the percentage of the data that contains anomalies. To evaluate the reliability of the anomaly-based IF machine learning algorithm, we set the “contamination” parameter on different ratios (0.00, 0.02, 0.04, 0.06, 0.08 and 0.1). The results of applying the IF algorithm utilizing various contamination ratios are depicted in Fig. 4.

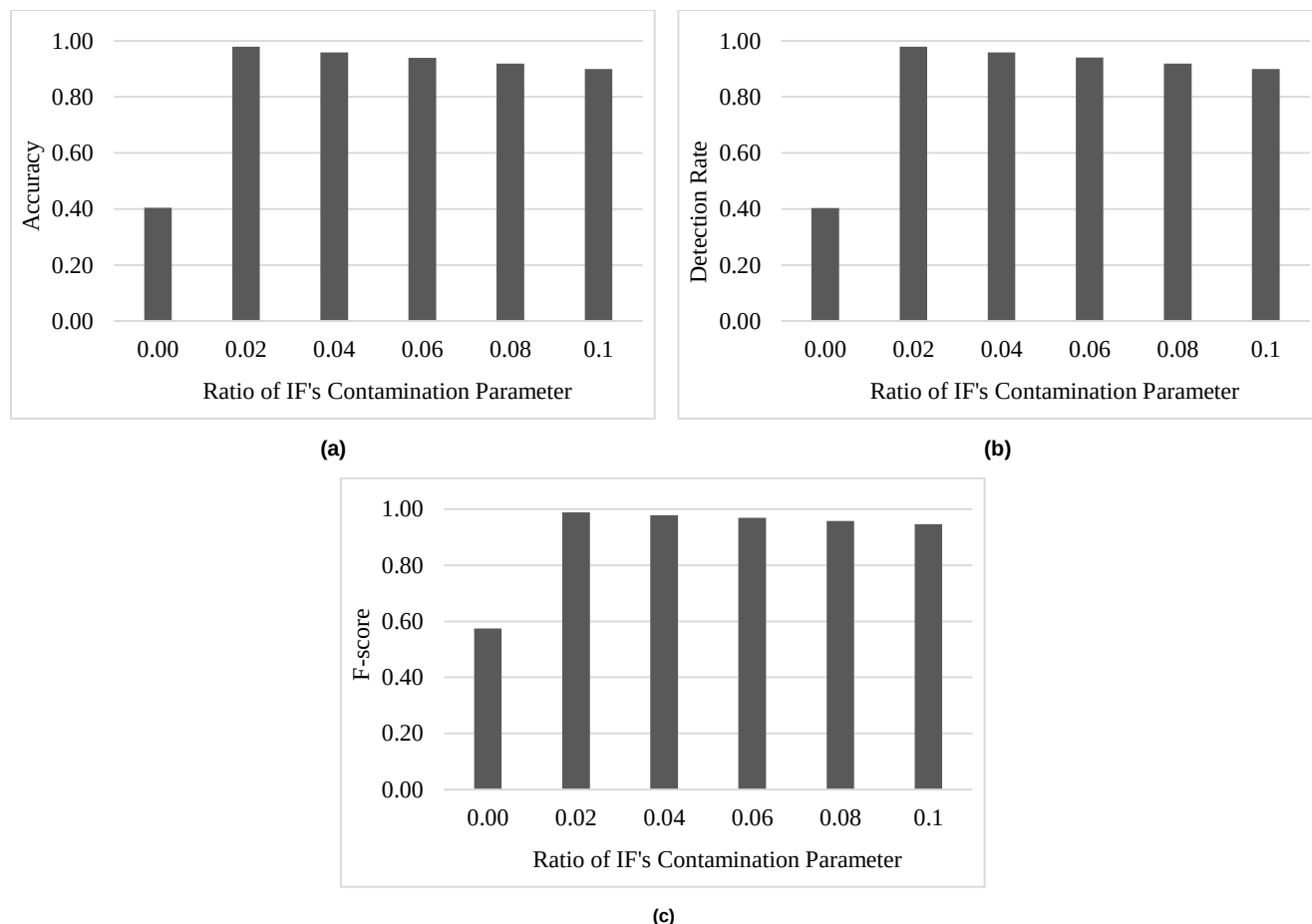


FIGURE 4. Subfigures (a), (b) and (c) are the accuracy, detection rate and F-score results of applying the IF algorithm with various contamination parameter ratios.

Figures 4a, 4b, and 4c show the results of applying the IF algorithm with three separate runs, each with a contamination parameter set to a different ratio from 0.0 to 0.1. When the contamination ratio is set to 0.00, the algorithm produces the worst results (40% accuracy, 40% detection rate, and 58% f-score). With a contamination ratio of 0.02, detection results are at their best (accuracy of 98%, detection rate of 98%, and f-score of 99%), followed by a ratio of 0.04 (accuracy and detection rate of 96%, and f-score of 99% respectively). We found that the minority class's (malicious instances') ratio in the dataset, or the true contamination ratio, is 0.03. The following outcomes are seen when the contamination ratio is

raised over the actual contamination ratio of the dataset. Accuracy (94%), detection (94%), and f-score (97%), and f-score (96%), respectively, are achieved with a contamination ratio of 0.06; accuracy (90%), detection (90%), and f-score (95%), respectively, are achieved with a ratio of 0.08; and accuracy (90%), detection (90%), and f-score (95%) are achieved with a ratio of 0.1. It is interesting to note that when the contamination ratio is close to the actual contamination ratio (between 0.02 and 0.04), the IF algorithm produces the best detection results. Table 6 provides a summary of the data collected.

TABLE 6
THE RESULTS OF APPLYING THE IF ALGORITHM UTILIZING VARIOUS CONTAMINATION RATIOS.

Contamination Ratio	Accuracy	Detection Rate	F-score
0.00	0.40	0.40	0.58
0.02	0.98	0.98	0.99
0.04	0.96	0.96	0.98
0.06	0.94	0.94	0.97
0.08	0.92	0.92	0.96
0.1	0.90	0.90	0.95

Table 6 provides the performance metrics scores that were obtained by applying the IF algorithm employing several runs while varying the contamination hyper-parameter of the IF algorithm and the parameter of the method somewhere between the range (0 - 1). It has been observed that when the contamination ratio is set to 0, the algorithm produces the worst possible results, with an accuracy and detection rate of 40%, and an f-score of 58%. On the other hand, the best detection results are achieved when the contamination ratio is set to 0.02, with an accuracy of 98%, a detection rate of 98%, and an f-score of 99%. We found that the true contamination ratio of the dataset is 0.03, which is the malicious observations ratio in the dataset.

It has been found that when the contamination parameter is adjusted to a level that is higher than the actual contamination ratio of the dataset by 0.02 gradual increasing of the contamination ratio, the detection performance of the IF algorithm is decreased by one and two percent over the three-performance metrics. Observably, when the contamination ratio is close to the real contamination ratio between the range (0.02 and 0.04), the use of the IF algorithm yields the greatest detection results. Thus, the tuning IF algorithm's parameter is something that should be taken into consideration.

The results of the anomaly-based Isolation Forest algorithm, when the contamination parameter is set to 0.02, are quite impressive with an accuracy of 0.98, a detection rate of 0.98, and an F-score of 0.99. Accuracy measures the proportion of correctly classified instances among all instances in the dataset. An accuracy of 0.98 indicates that the Isolation Forest algorithm is highly accurate in identifying

both normal and anomalous instances in the dataset. However, it's essential to consider the class distribution; if the dataset is highly imbalanced (e.g., only 2% of instances are anomalies), high accuracy alone may not be sufficient.

The detection rate, also known as true positive rate or recall, measures the proportion of actual anomalies correctly detected by the algorithm. A detection rate of 0.98 suggests that the Isolation Forest is excellent at identifying true insider threats while minimizing false negatives. This is a crucial metric for insider threat detection since failing to detect actual threats can have severe consequences.

The detection rate, also known as true positive rate or recall, measures the proportion of actual anomalies correctly detected by the algorithm. A detection rate of 0.98 suggests that the Isolation Forest is excellent at identifying true insider threats while minimizing false negatives. This is a crucial metric for insider threat detection since failing to detect actual threats can have severe consequences. In summary, the Isolation Forest algorithm with a contamination parameter of 0.02 achieves high accuracy, indicating its ability to classify both normal and anomalous instances correctly. The high detection rate suggests that it effectively identifies true insider threats, which is crucial for the security of the system. The exceptional F-score highlights the algorithm's balanced performance, making it a strong candidate for insider threat detection tasks. In subfigures (a), (b) and (c) of Fig. 5, we can see the improvement in performance of the anomaly-based IF machine learning algorithm compared to the four traditionally-applied supervised machine learning algorithms.

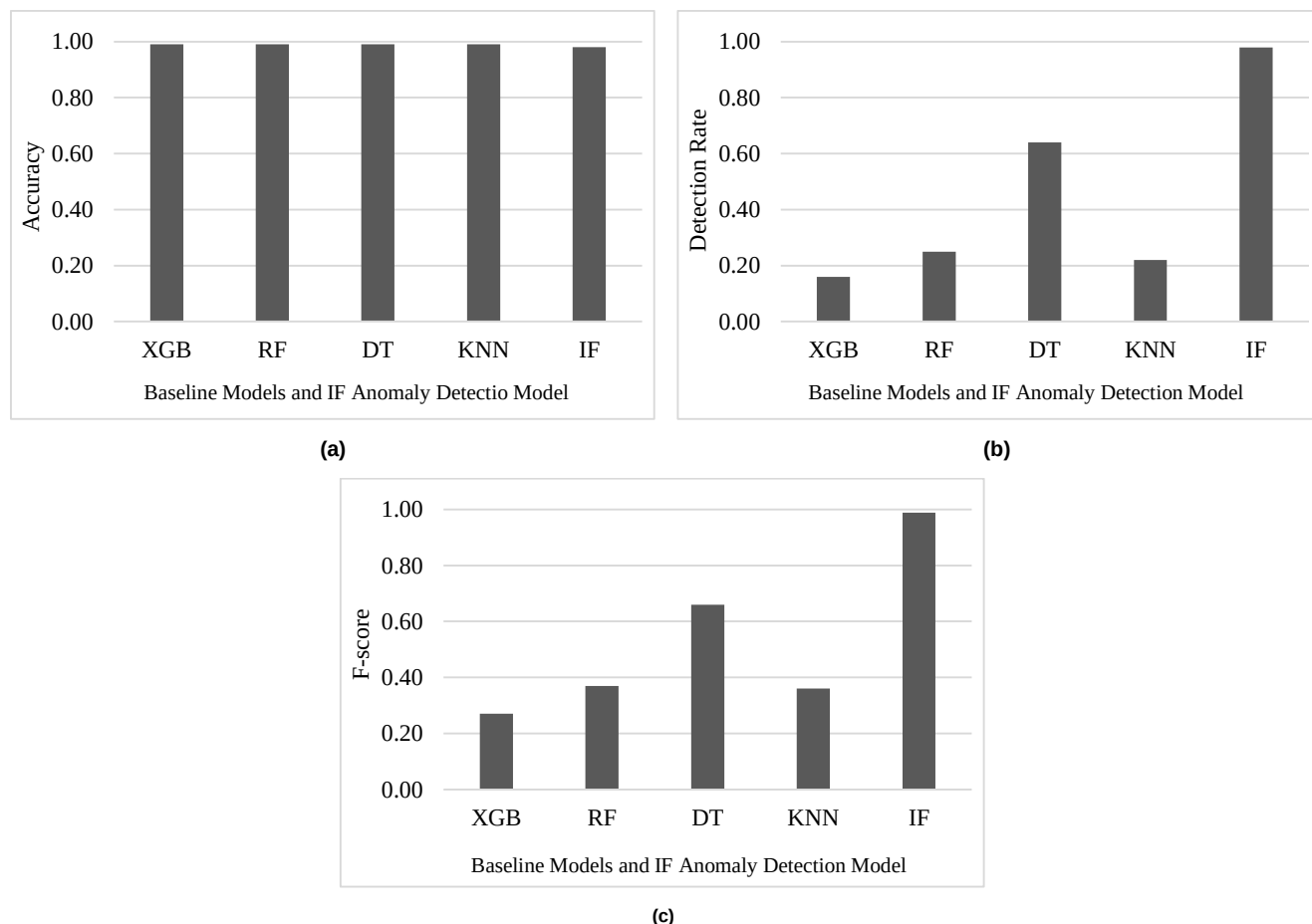


FIGURE 5. Subfigures (a), (b) and (c) results of the baseline Supervised Machine Learning Models compared with the IF Anomaly Detection Model.

We evaluate the suggested model against the standard and relevant studies to ensure its efficacy. Starting with the standard supervised machine learning algorithms as a baseline, we next test our anomaly-based IF machine learning algorithm's performance. The DT algorithm performs best in the baseline, with an accuracy of 99%, a detection rate of 64%, and an f-score of 66%; the RF and KNN algorithms perform similarly, but with lower detection rates and lower f-scores (25% and 37%, respectively). The XGB algorithm achieves the lowest detection result with an f-score of 27% and a detection rate of 16%.

The suggested approach is shown to greatly enhance the outcomes of insider threat detection when compared to the baseline. With a detection rate of 98% and an f-score of 99%, the anomaly-based IF machine learning algorithm outperforms the four conventional supervised machine learning algorithms used in the study of insider threats. Figure 5 demonstrates that when the dataset is highly skewed, the accuracy measure is almost the same across all applied techniques, concealing an accurate picture of the algorithm's performance. Instead, measurements like detection rate and f-score show how far ahead of the curve the IF anomaly-based machine learning system is compared to standard, supervised

methods. Because of its robustness against the significant imbalance problem of the dataset classes, the IF method is a promising alternative to conventional supervised machine learning techniques. Therefore, it has a high detection rate for malicious instances, even if such occurrences are uncommon in a given organization's system. Table 7 compares the results the proposed model with the existing models that are validated on the same dataset version (CERT r4.2).

TABLE 7
THE COMPARISON OF THE PROPOSED MODEL WITH THE EXISTING MODELS.

Ref.	Method	F-score
[49]	Autoencoder (AE)	92%
	Variational Autoencoder (VAE)	94%
[50]	Bio-Inspired models	70%
[51]	Multi fuzzy classifier (MFC)	82%
[21]	LR + SMOTE	67%
	NB + SMOTE	85%
	KNN + SMOTE	98%
Proposed model	IF	99%

As presented in Table 7, the comparison of our proposed study, which applies the Isolation Forest (IF) method with a 99% F-score, to existing works that employ different methods and achieve varying F-scores provides valuable insights into the effectiveness of each approach in insider threat detection. In [49], the AE and VAE methods achieved high F-scores of 92% and 94%, respectively, which indicates their effectiveness in capturing and detecting anomalies. Our proposed IF method outperforms both AE and VAE in terms of F-score. This suggests that IF may provide a simpler and more efficient approach for insider threat detection while maintaining high accuracy. In [50], the Bio-Inspired Method achieved an F-score of 70%, which is lower than our proposed IF method. IF demonstrates a substantial performance improvement compared to the bio-inspired method, highlighting its superiority in identifying insider threats in the given context. In [51], the MFC was applied and achieved an F-score of 82%, which is respectable but lower than the F-score achieved by IF in our proposed study. IF exhibits a higher F-score, suggesting that it may provide better discrimination between normal and malicious activities in the context of insider threat detection. In [21], the Logistic Regression (LR) + SMOTE, Naive Bayes (NB) + SMOTE, and K-Nearest Neighbors (KNN) + SMOTE methods were applied. These studies applied different machine learning algorithms in combination with the Synthetic Minority Over-Sampling Technique (SMOTE) to address class imbalance. While the F-scores achieved by LR + SMOTE, NB + SMOTE, and KNN + SMOTE are notable, our new proposed IF method surpasses all of them with a significantly higher F-score of 99%. The comparison suggests that IF, as an anomaly-based algorithm, is exceptionally effective in capturing insider threats, even without explicit resampling techniques like SMOTE.

In summary, our proposed study's application of the Isolation Forest (IF) method with a 99% F-score demonstrates its remarkable capability in insider threat detection. It outperforms existing works employing Autoencoders, Bio-

Inspired methods, Multi Fuzzy Classifiers, and various machine learning algorithms with SMOTE. The results suggest that IF is a powerful and efficient approach for identifying insider threats, with high precision and recall, making it a strong candidate for real-world insider threat detection applications.

V. CONCLUSION

The risk of an insider assault on the networked systems of private businesses and government institutions is high. One of the challenges of insider threat identification is that there is often very little ground truth available and the dataset is quite unequal. To improve the detection of harmful insider actions, this paper provides an anomaly-based detection strategy using IF algorithm to offset the effects of the highly imbalanced CERT r4.2 insider threat dataset. The research was conducted to better identify destructive insider actions. The proposed model is able to detect malicious insider behavior because it treats such acts as outliers relative to typical user activities, which make up the majority of observations in the dataset we use. The IF algorithm's contamination parameter is tweaked over the range of values (0.00, 0.02, 0.04, 0.06, 0.08, and 0.1) in order to find the best feasible model. After collecting data, we compare it to the norm and any applicable models. The results show that when the contamination parameter is set to 0.02, the suggested model performs optimally, with an accuracy and detection rate of 98% and an f-score of 99. When compared to competing models, ours shows considerable improvement in spotting insider threats and flexibility in adapting to new contexts. The anomaly-based IF machine learning approach has been shown to be useful for dealing with imbalanced datasets because of its capacity to spot outliers and irregularities. As a result of this reasoning, we conclude that the proposed model detects the insider threats with f-score of 99 while the dataset is extremely imbalanced. In summary, the Isolation Forest algorithm with a contamination parameter of 0.02 has demonstrated outstanding performance in insider threat detection. It's important to keep in mind that the choice of the contamination

parameter (0.02 in this case) may affect the results. Adjusting this parameter allows us to control the trade-off between precision and recall, depending on the specific requirements of the proposed insider threat detection system. Additionally, further evaluation, such as testing on a different dataset or exploring different contamination values, can help validate the algorithm's robustness and generalizability.

To further refine and improve model performance in further work, we will consider conducting a more comprehensive analysis, including hyper-parameter tuning, feature selection, and potentially exploring deep learning algorithms or ensemble techniques. Additionally, consider evaluating the models using additional metrics and assessing their performance in real-world scenarios to ensure practical utility in insider threat detection.

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